MIL-S-2580E
5 October 1979
SUPERSEDING
MIL-S-2580D
19 April 1971

MILITARY SPECIFICATION

SIGNBOARDS, BLANK (FOR TEMPORARY OUTDOOR SIGNS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers blank sheets for plain and reflective temporary outdoor signs.

1.2 Classification. The signboard sheets shall be of the following types and colors, as specified:

Type I - Plain
White (one side)

Type II - Reflective
White (one side)
Yellow (one side).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 9905

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin.
QQ-S-781	- Strapping, Steel and Seals.
TT-P-98	- Paint; Stencil, Flat.
UU-P-31	- Paper; General Specifications and Methods of Testing.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Box, Shipping Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-T-60	- Tape: Packaging, Waterproof.

MILITARY

MIL-P-130	- Paper, Wrapping, Laminated and Creped.

STANDARDS

FEDERAL

FED. TEST METHOD STD. NO. 141	- Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

U.S. BUREAU OF PUBLIC ROADS

Standard Yellow for Highway Signs and Markings

(Application for copies should be addressed to the Office of Highway Safety, Bureau of Public Roads, U.S. Department of Transportation, Washington, DC 20006.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

G26 - Operating Light and Water Exposure Apparatus, Xenon-Arc Type for Exposure of Nonmetalic Materials, Rec. Practice for.
D984 - Standard Method of Test for Reducible Sulfur in Paper.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Description. Signboards shall be sheets of glossy or reflective material which is receptive to paints used in screen process printing (see 6.5).

3.2 First article.

3.2.1 Preproduction model. The contractor shall furnish 20 signboards for examination and test within the time frame specified to prove, prior to starting production, that his production methods and choice of design detail will produce signboards that comply with the requirements of this specification. Examination and tests shall be as specificied in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Weathering. Signboards shall not curl, delaminate, or be otherwise damaged when subjected to 150 hours in a weatherometer with water cycle.

3.4 Curl. Signboards shall have no curl in excess of 3/16 inch in 16 inches in any direction.

3.5 Rigidity. The signboard shall not bend of its own weight more than 4-1/2 inches per foot in one direction; nor bend more than 8 inches per foot in the other direction nor shall the sum of these two deflections exceed 12 inches.

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3.6 Set resistance and flexibility. Signboards shall not be damaged nor take a permanent set when rolled into a 5-inch-diameter cylinder and held for 2 minutes.

3.7 Sizes and shapes. Signboards shall be square or round, as specified. Unless otherwise specified, square signboards shall be 16 inches each side and round shapes shall be 16 inches in diameter. A tolerance of plus or minus 1/4 inch shall be permitted. Special shapes, if required, shall be as specified (see 6.2).

3.8 Color

3.8.1 White (one side). One side of the signboard shall be white and the color of the other side shall be at the option of the supplier.

3.8.2 Yellow (one side). One side of the signboard shall be yellow and the color of the other side shall be at the option of the supplier. The yellow color (see 6.4) shall match the U.S. Bureau of Public Roads Standard Yellow for Highway Signs and Markings. The Munsell notation is 10YR 7.5/14 and the following limits shall apply:

	x	y	Y
Max.	.5117	.4613	.552
Min.	.4909	.4522	.489

3.9 Type I, plain. Type I signboards shall be of glossy material. The minimum physical properties shall be as follows:

Thickness	0.020 + 0.003 inches
Basis weight	75 + 12 lb./1,000 $ft.^2$
Tearing resistance, average (325 gm lowest for one direction)	400 gm, average of both directions
Bursting strength	120 points
Brightness (white only)	80 percent

3.10 Type II, reflective. Type II signboards shall be of reflective material which has enclosed or encapsulated lens and a smooth surface. The sulfur content of Type II paper or board signboards shall not exceed 0.02 percent. The minimum physical properties shall be as follows:

Thickness	0.022 $\pm$ 0.005 inches
Basis weight	105 $\pm$ 20 lb./1,000 $ft.^2$
Tearing resistance	400 gm, each direction
Bursting strength	160 points

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3.10.1 Reflection factor. Type II signboards shall have not less than the reflection factor values specified in Table I.

TABLE I. Minimum Reflectance Factor Values.

Divergence Angles (Degrees)	Incidence Angles (Degrees)				
	-4°	+10°	+20°	+30°	+50°
		White			
2	2.5	2.5	-	1.8	1.0
4	1.2	1.2	-	1.0	0.7
6	0.75	0.75	0.7	0.6	0.5
		Yellow			
2	2.0	2.0	-	0.97	0.43
4	0.97	0.99	-	0.62	0.28
6	0.61	0.61	0.58	0.45	0.23

3.10.2 Weathering. Type II signboards shall retain not less than 90 percent of the reflective factor values specified in table I after the weathering test specified in 4.6.7.

3.11 Workmanship. Signboards shall be uniform in construction and free fro pinholes, tears, cracks, cuts, creases, discolorations, delaminations, and any other defects that would affect the screen printing quality. The sheets shall not stick together so that force is required to separate them. Sheets shall be trimmed cleanly and neatly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. The inspection requirements are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.7).

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4.3 Preproduction inspection.

4.3.1 Examination. All preproduction signboards shall be examined as specified in 4.5.1. Presence of one or more defects on any signboard shall be cause for rejection.

4.3.2 Tests. All preproduction signboards shall be subject to the tests marked "X" in column 1 or 2, as applicable, of table II. Failure of any test on any signboard shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling

4.4.1.1 For examination. Sampling for examination shall be in accordance with MIL-STD-105, inspection level S-4.

4.4.1.2 For tests. Sampling for tests shall be in accordance with MIL-STD-105, inspection level S-1.

4.4.2 Examination. Samples selected in accordance with 4.4.1.1 shall be examined as specified in 4.5.1. AQL shall be 10 percent defective.

4.4.3 Tests. Samples selected in accordance with 4.4.1.2 shall be subjected to the tests marked "X" in column 3 or 4, as applicable, of table II. AQL shall be 10 percent defective.

4.5 Inspection procedure

4.5.1 Examination. Signboards shall be examined for the following defects:

101. Dimensions not as specified.
102. Signboard contains a curl in excess of 3/16 inch in 16 inches.
103. Signboards contain pinholes, tears, cracks, cuts, or discoloration.
104. Signboards stick together.
105. Signboards not trimmed cleanly and neatly.

4.5.2 Tests

4.5.2.1 Test conditions. Unless otherwise specified herein, test conditions shall be in accordance with UU-P-31, method 102.

4.5.2.2 Test schedule. Tests shall be conducted in accordance with table II.

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TABLE II. Test Schedule.

Prepro.		Qual. Conf.		Test	Test Para.	Req. Para.
Type I	Type II	Type I	Type II			
1	2	3	4	5	6	7
X	X	X	X	Thickness	4.6.1	3.9 or 3.10
X	X	-	-	Tearing resistance	4.6.1	3.9 or 3.10
X	X	-	-	Bursting strength	4.6.1	3.9 or 3.10
X	-	-	-	Brightness (white only)	4.6.1	3.9
X	X	X	X	Basis weight	4.6.2	3.9 or 3.10
X	X	X	X	Rigidity	4.6.3	3.5
X	X	X	X	Set resistance and flexibility	4.6.4	3.6
-	X	-	X	Visibility and reflectivity	4.6.6	Table I
-	X	-	X	Color tests (yellow only)	4.6.7	3.8.2
X	X	-	-	Screening and weathering	4.6.8	3.1, 3.[illegible] and 3.1[illegible]
X	-	X	-	Brightness	4.6.5	3.9
-	X	-	-	Sulfur content	4.6.9	3.10

4.6 Test procedure.

4.6.1 Physical properties. The following tests shall be conducted in accordance with UU-P-31:

Test	Method
Thickness	173
Tearing resistance	170
Bursting strength	112

Nonconformance to 3.9 or 3.10 as applicable, shall constitute failure of these tests.

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4.6.2 Basis weight. The conditioned samples shall be carefully weighed and computed to the weight per 1,000 square feet. Nonconformance to 3.9 and 3.10, as applicable, shall constitute failure of this test.

4.6.3 Rigidity. After conditioning, a 16-by-16-inch square sheet of signboard shall be laid on level drawing board or table having a square edge so that 12 inches of the signboard extends beyond the edge. A board covering all 4 inches of the signboard remaining on the table shall be clamped or weighted so as to hold the signboard. Two straight edges shall be laid on the table, one on either side of the signboard, projecting 12 inches beyond the edge of the table, in such a manner as to project the plane of the table's surface. The other edge of the signboard shall be allowed to hang free. Signboards used for this test shall not have any curl, shall lay flat when placed on a flat surface, and shall not be allowed to hang free for more than 1 minute during this test. When the edge has come to rest, vertical measurements shall be taken from each outer corner to the straightedge above it, and averaged. The signboard shall then be turned 90 degrees, turned over, and remeasured. Averaged readings for one position exceeding 4-1/2 inches, or averaged readings for the other position exceeding 8 inches, or the sum of the two averaged readings exceeding 12 inches shall constitute failure of this test.

4.6.4 Set resistance and flexibility. A 16- by 16- inch square sheet of signboard (not one that has been used in previous tests) shall be bent into a cylindrical shape so that two edges butt, and shall be held in this position for not less than 2 minutes. A 5-inch-diameter mandrel may be used for this purpose. The sheet shall be immediately laid on a flat surface so that the convex side is up and the greatest separation between signboard and surface shall be measured 55 to 60 seconds after unrolling. Any measurement in excess of 3/16 inch shall constitute failure of this test. The foregoing procedure shall then be repeated with the other two edges of the sheet, with the other side out. After each measurement, the signboard shall be examined. Any visible evidence of damage such as cracking, spalling, or delamination shall constitute failure of this test.

4.6.5 Brightness. The Type I sample brightness shall be determined by method 4251 or method 4252 of FEDERAL TEST METHOD STANDARD No. 141. When method 4251 is used and the spectrophotometer is equipped with a properly operating tristimulus integrator, the "Y" value (brightness) can be used instead of the computations as required by the method. The inability of the samples to equal or exceed the brightness value in 3.9 shall constitute failure of the test.

4.6.6 Visibility and reflectivity (type II only)

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4.6.6.1 Apparatus. Arrangement of the apparatus for the reflectivity test shall be as shown in figure 1. A light projector having a lens diameter of not more than 1 inch and capable of projecting a uniform field of light shall be used to illuminate the samples. The lamp shall have a color temperature of not less than 2850°K. The light meter shall have a photoelectric receiver whose response has been corrected with a Kodak Wratten Filter No. 102, or equal. The maximum dimension of the active area of the receiver shall be 1 by 2 inches. The long dimension of the cell shall be perpendicular to the radial line from the center of the projector lens (perpendicular to the beam of light). Samples shall be mounted on a flat-black (test) surface not less than 3 feet square which, when tested without any sample, shall give no appreciable reading. The sample shall be 50 feet plus or minus 2 inches from the projector lens and the receiver. The projector shall be adjusted, or masked, or both, so that the incident light falls entirely on the sample.

4.6.6.2 Sample. Samples shall be 12 plus or minus 1/8 inch square, or a 13-1/2 plus or minus 1/8 inch circle. Any masking shall be of flat-black material.

4.6.6.3 Procedure. With the equipment set in a darkened room, turn on the projector, aline the receiver toward the black test surface, and without any sample, zero the meter. Mount a sample on the test surface and adjust the apparatus to the appropriate divergence and incidence angles specified in table I. Measure the distance from the projector to the sample, the area of the sample, and the illumination incident on the sample. Measure the illumination incident on the receiver due to the reflection from the sample at each angle of incidence for each angle of divergence. The angles of incidence and divergence shall be as specified in table I. L_r shall be measured in the same units as L_s. Compute the reflective intensity (R) as follows:

$$R = \frac{L_r(d^2)}{L_s(A)}$$

L_r = illumination incident upon the receiver.
L^r = illumination incident upon the sample.
d^s = distance in feet from sample to projector.
A = area in square feet of the sample.

Inability to equal or exceed the reflective intensity readings specified in table I shall constitute failure of this test.

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4.6.7 Color tests. Type II samples shall be compared with the current "Standard Yellow for Highway Signs and Marking" of the U.S. Bureau of Public Roads by method 4250 of FED TEST METHOD STD. No. 141. Observers shall have normal color vision; if males, they shall have passed a standard color chart test. Failure to fall within the tolerance limits of this standard shall constitute failure of this test. If there is any disagreement, samples shall be tested by method 4252 of FED. TEST METHOD STD. No. 141. Chromaticity coordinates and luminance factor failing to fall within the ranges specified in 3.8.2 shall constitute failure of this test.

4.6.8 Screening and accelerated weathering. Preproduction samples shall be screen stenciled or printed with black stencil paint conforming to TT-P-98, type II. Screening shall be distributed over the face of the entire sample in a pattern that will cover more than one-third but not more than three-quarters of each sample. After the paint has dried, samples shall be examined. Any area which does not take ink evenly shall constitute failure of this test. Samples shall then be subjected to accelerated weathering in accordance with test method 6151 or 6152 of FED. TEST METHOD STD. No. 141 or ASTM G-26 Type BH for not less than 150 hours, after which they shall be examined and the reflective intensity measured as specified in 4.6.5. Any evidence of damage, delamination or discoloration or nonconformance to 3.10.2 shall constitute failure of this test.

4.6.9 Sulfur content. Any paper product in type II signboards shall be tested in accordance with ASTM D984. Any sulfur percentage greater than 0.02 percent by weight of the air-dry paper shall constitute failure of this test.

4.7 Inspection of Packaging

4.7.1 Quality conformance inspection of pack

4.7.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.7.1.3 Examination. Samples selected in accordance with 4.7.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

106.	Materials or containers not as specified for level A or B. Each incorrect material or each incorrect container shall be considered one defect.
107.	Signboards not preserved in quantities of 100.

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108. Strapping not as specified.
109. Gross weight or size exceeds the size limitations of the boxes for level B.
110. Marking illegible, incomplete, incorrect, or missing.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, level B, or Commercial as specified (see 6.2).

5.1.1 Level A. The signboards of like type, size and color shall be neatly stacked in quantities of 100. Each stack of 100 sheets shall be wrapped with paper conforming to MIL-Q-130, type III, or with plastic conforming to L-P-378, type I. The wrap shall be secured with tape conforming to PPP-T-60 or shall be waterproof sealed by thermosealing process. Each stack shall then be preserved in a close-fitting box conforming to PPP-B-636, W5c or V3c. Box closure shall be in accordance with the appendix to the box specification, method V.

5.1.2 Level B. The signboards shall be stacked, wrapped, and preserved as specified in 5.1.1, except the box shall be class domestic, type CF, variety SW, grade 175, and box closure shall not be required to be method V.

5.1.3 Commercial. The signboards of like type, size, and color shall be preserved, in quantities of 100, in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, level B, or Commercial as specified (see 6.2).

5.2.1 Level A. The signboards, preserved as specified in 5.1 shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type, style optional. Strapping shall conform to QQ-S-781, class I, type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2), strapping shall be finish A.

5.2.2 Level B. The signboards, preserved as specified in 5.1 shall be packed in close-fitting boxes conforming to PPP-B-636, V3c, style optional, or in boxes conforming to PPP-B-640, class 2, style optional. The gross weight or size shall not exceed the limitations of the applicable box specification.

5.2.3 Commercial. The signboards, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

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5.3 Marking

5.3.1 Military Packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial Packaging. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. Signboards are intended for use as the base for screen processed signs which will be exposed to the weather for periods not to exceed 90 days. Type I is for general use, type II is for use at night where sufficient illumination exists to make their added reflectivity effective. White (one side) is for general use and when yellow is required, the white can be painted yellow of paragraph 3.8.2.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and color required (see 1.2).
(c) Time frame required for submission of preproduction model (see 3.2.1).
(d) Whether signboards shall be square or round; and the size required if other than as specified in 3.7.
(e) Size and shape if special shapes are required (see 3.7).
(f) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(g) When finish A strapping is required (see 5.2.1).

6.3 Preproduction model. Any changes or deviations of production signboards from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish signboards conforming to this specification.

6.4 PBR highway yellow. The vivid orange-yellow color specified herein falls between ISCC-NBS Centroid Colors No. 66 and No. 68 and is equivalent to chip No. 13538 of FED. STD. No. 595.

6.5. Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.1).

Custodians
Army - ME
Air Force - 99

User activities
Army - GL
Navy - MC

Preparing activity
Army - ME

Project No. 9905-0244

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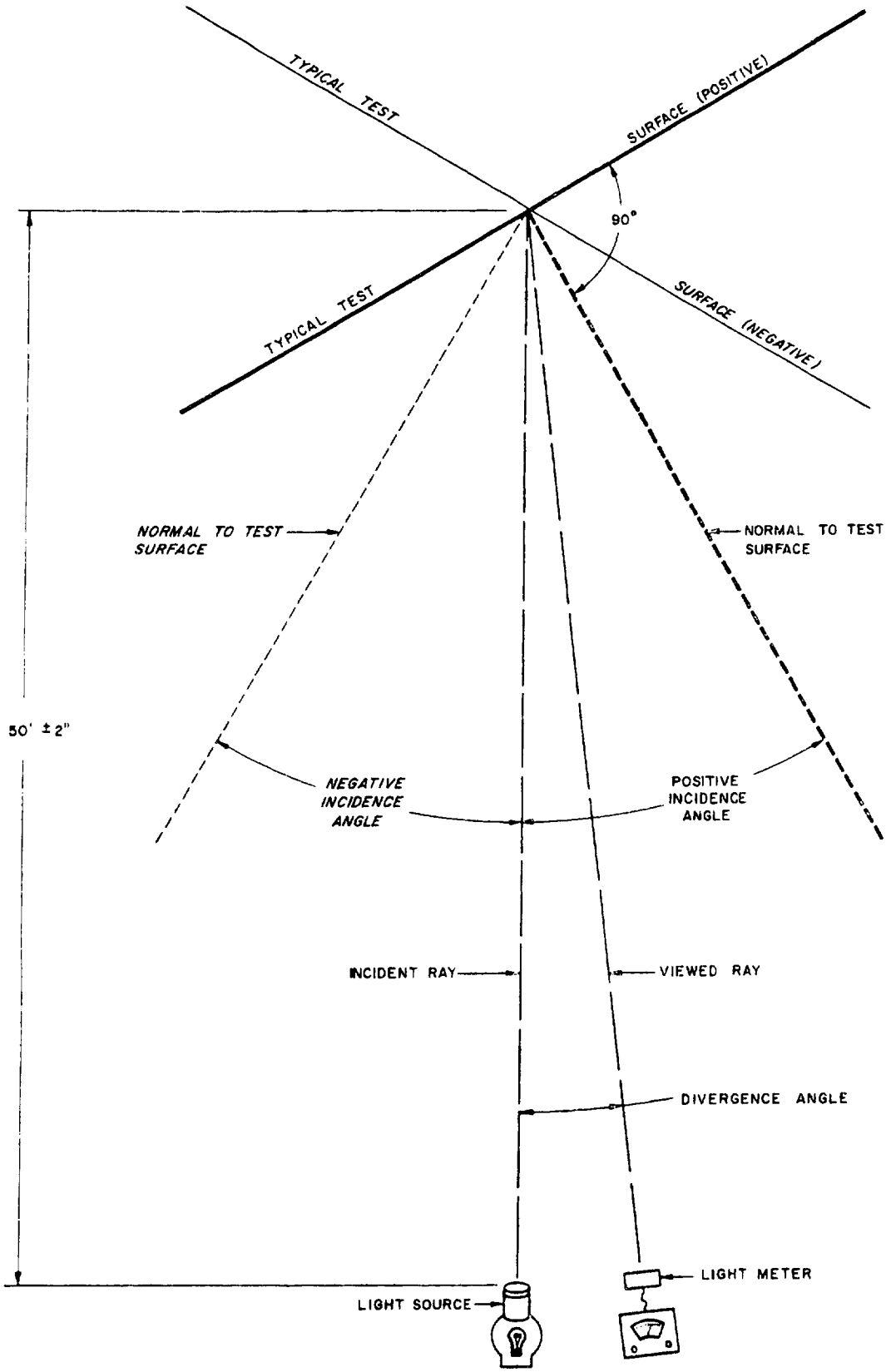


FIGURE 1 REFLECTIVITY TEST

DX210

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-S-2580E Signboards, Blank (for Temporary Outdoor Signs)

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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